

# What Do Lorentz-Equivariant Jet Taggers Learn?

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## Abstract

Lorentz-equivariant networks are increasingly important for collider physics, but little is known about the physical information they represent internally. We present, to our knowledge, the first interpretability study of the Lorentz Geometric Algebra Transformer (L-GATr), combining equivariance tests, linear probes, grade ablations, and geometric-algebra-invariant probes on TopTagging and JetClass. After first verifying that the trained model behaves equivariantly under boosts, we find that its scalar channels encode jet mass and N-subjettiness while suppressing frame-dependent  $\eta$  information, and that its multivector grades show structured usage: grade-2 bivectors are the most sensitive TopTagging ablation channel, while preliminary JetClass ablations suggest a shift toward grade-3/4 importance. These results suggest that geometric-algebra architectures learn physically meaningful internal representations. More broadly, these tools may help validate learned symmetries in future collider foundation models.

## 1. Introduction

Equivariant neural networks are becoming central tools for scientific machine learning because they build known physical symmetries directly into the model. In high-energy physics (HEP), Lorentz symmetry is especially important: jets are observed in a particular detector frame, but the underlying physics is constrained by relativistic transformations. Recent Lorentz-equivariant architectures achieve strong performance on jet tagging benchmarks, yet high accuracy alone does not tell us what physics these models have learned, whether their symmetry constraints are active in practice, or how their internal representations differ from ordinary transformers.

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This question is timely for trustworthy AI in science. Collider analyses increasingly rely on neural networks to extract subtle structure from complex events. If symmetry-aware models are to become reliable scientific instruments, we need tools that can inspect not only their outputs, but also their internal organization. Interpretability for equivariant models is therefore not only a debugging exercise: it is a way to ask whether neural networks discover representations that align with the structure of the physical problem.

We study the Lorentz Geometric Algebra Transformer (L-GATr (Spinner et al., 2024)), which builds on the Geometric Algebra Transformer framework (Brehmer et al., 2023) by representing jet constituents as multivectors in the geometric algebra of Minkowski spacetime. This representation decomposes into grades: scalars, vectors, bivectors, trivectors, and pseudoscalars. The grade structure gives L-GATr an unusual advantage for interpretability: unlike a generic hidden vector, each component has a transformation law and a geometric meaning. This makes L-GATr a useful testbed for asking how symmetry-aware scientific models organize information.

We study L-GATr using equivariance checks, linear probes, grade ablations, and invariant multivector probes. The Lorentz boost experiment serves as an important validation step: it checks that the trained network realizes the equivariance promised by the architecture, and provides a baseline for comparing equivariant and non-equivariant representations. Our main contributions are:

**(1) Linear probes reveal invariant physical representations.** Across 13 jet- and particle-level targets, scalar-channel probes show strong encoding of jet mass and N-subjettiness, while pseudorapidity  $\eta$  remains approximately absent from equivariant scalar representations. Non-equivariant baselines retain nonzero  $\eta$  information, providing a representation-level signature of the effect of Lorentz equivariance.

**(2) Grade ablations and invariant probes expose structured multivector usage.** We combine grade-wise ablations with 832 geometric-algebra-invariant features constructed from the full multivector representation  $h_{mv}$ . The results suggest a staged representational pattern on TopTagging: grade-2 bivectors are most important in early and global ablations, mid-layer probes show information distributed

across grade interactions, and the final classifier reads grade-0 scalar information by architectural design.

**(3) Preliminary JetClass results suggest task-dependent grade usage.** On TopTagging, grade-2 bivectors are the most sensitive ablation channel. On the 10-class JetClass benchmark, preliminary ablations suggest a shift toward grade-3/4 importance. We view this as exploratory evidence, with additional JetClass experiments ongoing, that L-GATr may not use a fixed grade recipe across tasks.

**(4) We introduce a symmetry-aware interpretability workflow.** Equivariance checks, invariant probes, and grade interventions together provide a practical toolkit for studying scientific models whose hidden states carry structured group representations. Such tools may help validate learned symmetries in future collider foundation models.

**Relation to prior work.** Prior interpretability studies of jet neural networks have focused mainly on attention maps for non-equivariant architectures. Wang et al. (2024) study ParT attention, finding sparsity and subjet structure, while Legge et al. (2025) caution that attention maps alone are incomplete explanations. Esmail et al. (2026) use attention maps and CKA to argue for physics-meaningful representations in IAFormer. Linear probes were introduced to jet-tagging interpretability by Cheng (2019), who found that individual N-subjettiness values are more accessible than their ratios. We extend this probe-based approach to transformer architectures and Lorentz-equivariant models, and add grade-aware interventions that are specific to geometric-algebra networks.

Additional analyses, including attention maps, CKA layer similarity (Kornblith et al., 2019), expanded multivector probe trajectories, and bootstrap uncertainty quantification, are reported in the Appendix. Our goal is not a complete mechanistic account of L-GATr, but a practical interpretability toolkit that reveals physically meaningful structure.

## 2. Background

### 2.1. Geometric Algebra and the Lorentz Group

The geometric algebra over Minkowski spacetime, with metric signature  $(+, -, -, -)$ , is a 16-dimensional associative algebra generated by four basis vectors  $e_0, e_1, e_2, e_3$  satisfying  $e_0^2 = +1, e_i^2 = -1$  for  $i = 1, 2, 3$ , and  $e_\mu e_\nu = -e_\nu e_\mu$  for  $\mu \neq \nu$ . Elements are called *multivectors* and decompose into five *grades*:

- **Grade 0** (scalar, 1 component): Lorentz-invariant by construction.
- **Grade 1** (vectors, 4 components): transform as 4-vectors under  $\text{SO}^+(1, 3)$ .
- **Grade 2** (bivectors, 6 components): span the Lie algebra

$\mathfrak{so}(1, 3)$ ; split into boost planes  $e_{0i}$  (3 components) and rotation planes  $e_{ij}$  (3 components).

- **Grade 3** (trivectors, 4 components): Hodge-dual to grade-1 vectors via the pseudoscalar  $I$ .
- **Grade 4** (pseudoscalar, 1 component): parity-odd invariant.

The grade-2 subspace is particularly significant: it spans the Lie algebra  $\mathfrak{so}(1, 3)$ , whose six generators correspond to three boost planes and three rotation planes. The geometric product of two multivectors mixes grades in a way determined by the algebra; for instance, the product of two grade-2 elements produces grades 0 and 4, while a grade-2 times a grade-1 element produces grades 1 and 3. This grade-mixing structure provides one possible lens for interpreting the empirical layerwise patterns described in Section 5.

### 2.2. Lorentz Geometric Algebra Transformer

L-GATr (Spinner et al., 2024) adapts the Geometric Algebra Transformer framework (Brehmer et al., 2023) from Euclidean to Minkowski spacetime. It represents each jet constituent as a multivector token encoding its 4-momentum, supplemented by beam spurion tokens (a lightlike vector along the beam axis and a time reference) that break the remaining boost symmetry as a fixed reference frame. The network consists of  $N_{\text{blocks}}$  GATrBlocks, each applying equilinear transformations (linear maps that commute with the Lorentz group action) and geometric attention over multivector tokens. Each block produces multivector channels  $h_{\text{mv}} \in \mathbb{R}^{N \times C_{\text{mv}} \times 16}$  and scalar channels  $h_s \in \mathbb{R}^{N \times C_s}$ .

The output layer `linear_out` is equilinear: it maps grade- $g$  inputs to grade- $g$  outputs only, with no grade mixing. Since the classification logit is a scalar (grade-0), `linear_out` reads *only* from the grade-0 component of  $h_{\text{mv}}$  at the final layer. This architectural constraint implies that all non-grade-0 components at the final layer are architecturally inaccessible to the output, regardless of their information content—a fact we use explicitly in interpreting the layer-resolved ablations.

The scalar channels  $h_s$  are Lorentz-invariant by construction: because the architecture is equivariant and the scalar representation of the Lorentz group is trivial,  $h_s(\Lambda x) = h_s(x)$  for any Lorentz transformation  $\Lambda$ .

L-GATr-slim (Petitjean et al., 2025) is a compact variant that retains only grade-0 scalars and grade-1 4-vectors internally, explicitly dropping the outer product and grade-2/3/4 channels. It has scalar channels  $h_s \in \mathbb{R}^{N \times 96}$  and vector channels  $h_v \in \mathbb{R}^{N \times 32 \times 4}$ , matching L-GATr’s task AUC while being  $6\times$  faster (Petitjean et al., 2025). Our grade decomposition results provide a direct empirical test of whether the higher-order grades dropped by L-GATr-slim carry task-relevant information.

### 3. Experimental Setup

#### 3.1. Datasets

**TopTagging** (Butter et al., 2018; Kasieczka et al., 2019) is a binary jet tagging benchmark with 1.2M training, 400k validation, and 400k test jets at  $\sqrt{s} = 14$  TeV. Signal jets arise from hadronic top quark decays  $t \rightarrow Wb \rightarrow qqb$  (three-pronged substructure); background jets are QCD-initiated single-prong jets. Constituent particles are ordered by transverse momentum  $p_T$  (descending).

**JetClass** (Qu et al., 2022) is a 10-class jet identification benchmark comprising jets from:  $Z \rightarrow \nu\nu$ ,  $H \rightarrow bb$ ,  $H \rightarrow cc$ ,  $H \rightarrow gg$ ,  $H \rightarrow WW \rightarrow 4q$ ,  $H \rightarrow WW \rightarrow 2q1\ell$ ,  $t \rightarrow Wb$  (hadronic),  $t \rightarrow Wb$  (leptonic),  $W \rightarrow qq$ , and  $Z \rightarrow qq$  decays, spanning a broad range of jet prong multiplicities and decay kinematics.

#### 3.2. Models

We study four models on TopTagging. **L-GATr** (Spinner et al., 2024): 12 GATrBlock layers,  $C_{mv} = 16$ ,  $C_s = 32$ , test AUC  $\approx 0.986$ . **L-GATr-slim** (Petitjean et al., 2025): 12 LGATrSlimBlock layers,  $C_v = 32$ ,  $C_s = 96$ , test AUC  $\approx 0.986$ . **ParT** (Qu et al., 2022): ParticleTransformer, kinematics-only checkpoint, 9 layers including embedding, test AUC = 0.9806. **Vanilla Transformer**: a standard transformer without Lorentz equivariance, using the same general architecture family as L-GATr but without symmetry enforcement, 10 blocks, test AUC = 0.9851. The vanilla transformer serves as a non-equivariant architectural reference rather than a perfectly matched control. For JetClass, we study L-GATr (OvO AUC = 0.98839) and L-GATr-slim.

#### 3.3. Probe Methods

**Equivariance test.** For each model, we apply 5 random Lorentz boosts to 200 test jets and measure the mean relative change in the output logit. We additionally perform a boost sweep with  $\gamma \in \{1.5, 2, 5, 10, 50\}$  to probe behavior at extreme boosts. Concretely, the output metric is  $|\ell(\Lambda x) - \ell(x)|/(|\ell(x)| + \varepsilon)$  with  $\varepsilon = 10^{-8}$ .

**Linear probes on  $h_s$ .** We extract scalar channel representations at each layer for 10,000 TopTagging jets, then fit linear probes for 13 physics targets: jet mass, jet multiplicity, N-subjettiness values  $\tau_1, \tau_2, \tau_3, \tau_{21}$  (Thaler & Van Tilburg, 2011), top/QCD classification, and particle-level  $p_T, \eta, \phi, E, p_T$ -quartile, and  $\Delta R$ . For L-GATr and L-GATr-slim, we report bootstrap confidence intervals based on  $B = 1000$  resamples with an out-of-bag (OOB) evaluation design: for each resample, probes are fit on the in-bag jets and evaluated on the corresponding OOB jets, so the reported scores are held-out. Regression targets use Ridge regression; binary and multi-class targets use logistic regression. For particle-

Table 1. Logit change under random Lorentz boosts (mean over 200 jets  $\times$  5 transforms). Lower is more equivariant.

| MODEL       | LOGIT CHANGE    | VS. L-GATr    |
|-------------|-----------------|---------------|
| L-GATr      | 5.61E-3 (0.56%) | 1 $\times$    |
| L-GATr-SLIM | 6.56E-4 (0.07%) | 0.12 $\times$ |
| ParT        | 1.15 (115%)     | 205 $\times$  |
| VANILLA-T   | 1.27 (127%)     | 226 $\times$  |

level probes, train/test separation is performed at the jet level to avoid particle leakage across splits. Additional implementation details are given in Appendix C.

**Grade decomposition.** We zero one grade of  $h_{mv}$  at all layers simultaneously and measure the AUC drop (*zero-grade ablation*), and separately activate only one grade while zeroing all others (*keep-only ablation*). Layer-resolved ablation zeros one grade at one layer at a time, all other layers intact. Bootstrap significance testing ( $B = 1000$ , all p-values  $< 0.001$ ) is reported for TopTagging. These interventions are applied to hidden multivector outputs after the corresponding layer module, not to the raw input embedding.

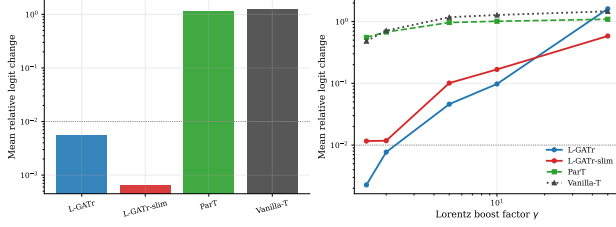
**Geometric-algebra-invariant probes on  $h_{mv}$ .** We construct 832 Lorentz-invariant scalar features per token from  $h_{mv}$  via the geometric-algebra inner product  $\langle \tilde{X}Y \rangle_0$  across grade pairs, where  $\tilde{\cdot}$  denotes reversion. Features include: grade-0/4 direct components (32 features), grade-1 pairwise inner products (136), grade-3 pairwise inner products (136), grade-1  $\times$  grade-3 cross terms (256), grade-2 scalar invariants  $\langle B_i B_j \rangle_0$  (136), and grade-2 pseudoscalar invariants  $\langle B_i B_j \rangle_4$  (136). These features are Lorentz-invariant by construction, enabling probes of the full multivector without introducing direct frame-dependent coordinates into the probe features. We give the exact feature grouping and evaluation protocol for these probes in Appendix C.

## 4. Validation and Physical Encoding

### 4.1. Equivariance Check

Table 1 reports the mean relative change in output logit under random Lorentz boosts for all four models. L-GATr achieves 0.56% logit error—consistent with floating-point accumulation across 12 layers—while ParT and the vanilla transformer exhibit errors of 115% and 127% respectively, approximately 200 times larger. We treat this experiment as validation rather than a main result: it confirms that the trained model and numerical implementation preserve the intended symmetry closely enough for the representation-level analyses below.

The boost sweep in Figure 1 shows that L-GATr maintains below 1% logit error for  $\gamma \lesssim 2$ , which covers the physically relevant range for LHC top jet production. Errors



**Figure 1.** Equivariance quantification. **Left:** mean logit change under random Lorentz boosts; both equivariant models remain near zero, with L-GATr at 0.56%, L-GATr-slim at 0.07%, and the non-equivariant baselines above 100%. **Right:** pure-boost sweep of mean relative logit change versus  $\gamma$  for all four models. The equivariant models stay far more stable across the sweep, whereas ParT and Vanilla-T show large logit changes already at  $\gamma = 1.5$  and remain highly non-invariant thereafter. At very large  $\gamma$ , the equivariant models also degrade, which we attribute to finite-precision numerical effects under extreme boosts rather than a breakdown in the moderate-boost regime relevant to TopTagging.

grow at larger  $\gamma$  values, consistent with floating-point precision degradation at extreme boosts rather than a fundamental failure of equivariance. The right panel extends this comparison to a controlled pure-boost sweep over  $\gamma \in \{1.5, 2, 5, 10, 50\}$  for all four models. The non-equivariant baselines degrade sharply even at moderate boosts, while the two equivariant models remain substantially more stable. At the largest boost factors, however, both equivariant models also show noticeable output-level degradation; for the full L-GATr, the logit-change metric at  $\gamma = 50$  even exceeds the corresponding ParT/vanilla values. We interpret this as an extreme-boost numerical robustness effect in finite precision, rather than evidence against the symmetry constraint in the physically relevant regime.

## 4.2. Scalar Channels Encode Physics Differentially Across Models

Table 2 reports linear probe scores at the final layer for all four models. Several consistent patterns emerge.

**Pseudorapidity  $\eta$  as an invariance probe.** For both equivariant models, the particle-level  $\eta$  probe score is statistically consistent with zero throughout all 12 layers (final layer:  $R^2 = 0.001$  and  $-0.000$  respectively, 95% CI  $[-0.003, 0.003]$  and  $[-0.007, 0.005]$ ). In contrast, ParT maintains  $R^2 = 0.22$  and the vanilla transformer  $R^2 = 0.10$  at their final layers. The vanilla transformer’s  $\eta$  probe *peaks* at layer 2 ( $R^2 = 0.627$ ) before decaying to 0.10—consistent with a non-equivariant network partially suppressing frame-dependent information, but being unable to eliminate it without an architectural constraint. The particle-level  $\phi$  probe is near zero for all models ( $R^2 \approx 0$ ), consistent with the approximate azimuthal symmetry of the dataset. Together, these results are consistent with the scalar channels suppressing linearly accessible frame-dependent rapidity

information. We treat this as a useful invariance sanity check for  $h_s$ , rather than as a complete statement that every internal quantity in the model is free of spurion-mediated lab-frame correlations.

**N-subjettiness and the  $\tau_{21}$  gap.** All models encode individual N-subjettiness values  $\tau_1, \tau_2, \tau_3$  strongly at the final layer ( $R^2 \approx 0.85\text{--}0.98$ ), consistent with these being well-defined jet substructure observables (Thaler & Van Tilburg, 2011). However, the ratio  $\tau_{21} = \tau_2/\tau_1$  is substantially weaker for all models ( $R^2 \approx 0.49\text{--}0.64$ ). This is consistent with the observation of Cheng (2019) that neural networks encode the individual  $\tau_N$  building blocks rather than directly representing their ratios—a pattern that appears to persist across very different architectures and training regimes. Our results extend this finding to Lorentz-equivariant transformers and ParT, suggesting it may reflect a general property of how the HEP loss landscape organises representations rather than an artefact of any single architecture.

**Full L-GATr encodes physics more richly than L-GATr-slim.** Despite identical task AUC (0.9855 both), L-GATr achieves higher final-layer probe scores for jet mass ( $R^2 = 0.980$  vs. 0.950), jet multiplicity (0.932 vs. 0.831),  $\tau_3$  (0.831 vs. 0.747), and  $\tau_2$  (0.890 vs. 0.854), all with non-overlapping 95% confidence intervals. This suggests that the richer multivector representation of L-GATr encodes a broader range of physical information even when downstream task performance is the same—consistent with L-GATr-slim’s design choice to drop the higher-order grades carrying geometric information.

**Particle  $p_T$  decays with depth in L-GATr.** The  $p_T$  probe peaks at layer 4 ( $R^2 = 0.946$ ) and decays to 0.714 at layer 12. This pattern, which is not observed as strongly in the non-equivariant models, may reflect the progressive abstraction of raw kinematics into higher-level jet structure representations as network depth increases.

**L-GATr-slim self-corrects input leakage.** L-GATr-slim takes 8 scalar input features including  $\log p_T$ ,  $\log E$ , and  $\Delta\eta$  per particle, resulting in an  $\eta$  probe score of 0.046 at layer 0. This decays monotonically to 0.000 by layer 12, suggesting that the Lorentz equivariance constraint progressively removes non-invariant information introduced at the input.

Figure 2 complements the final-layer table with a compact layerwise view of four representative probe targets. In particular, it makes the  $\eta$  suppression result visually immediate: the equivariant models stay near zero throughout depth, while ParT and Vanilla-T retain substantial non-zero  $\eta$  signal.



Table 2. Linear probe scores at the final layer for all four models. For L-GATr and L-GATr-slim, 95% bootstrap confidence intervals ( $B = 1000$ , OOB design) are reported. **Part**  $\eta$  row highlighted:  $\approx 0$  for equivariant models (consistent with suppression of linearly accessible frame-dependent rapidity information in  $h_s$ ), non-zero for non-equivariant models. Dash: not computed.

| Probe                         | L-GATr L12 [95% CI]        | L-GATr-slim L12 [95% CI]    | ParT L8      | Vanilla-T L10 | Metric |
|-------------------------------|----------------------------|-----------------------------|--------------|---------------|--------|
| top/QCD                       | 0.986 [.984, .988]         | 0.986 [.984, .988]          | 0.981        | 0.983         | AUC    |
| jet mass                      | 0.980 [.979, .981]         | 0.950 [.947, .954]          | 0.987        | 0.985         | $R^2$  |
| jet mult                      | 0.932 [.929, .935]         | 0.831 [.821, .840]          | 0.858        | 0.917         | $R^2$  |
| jet $\tau_1$                  | 0.952 [.950, .954]         | 0.930 [.926, .934]          | 0.976        | 0.963         | $R^2$  |
| jet $\tau_2$                  | 0.890 [.885, .895]         | 0.854 [.846, .861]          | 0.919        | 0.898         | $R^2$  |
| jet $\tau_3$                  | 0.831 [.823, .838]         | 0.747 [.735, .758]          | 0.852        | 0.788         | $R^2$  |
| jet $\tau_{21}$               | 0.519 [.497, .537]         | 0.488 [.469, .505]          | 0.641        | 0.525         | $R^2$  |
| part $p_T$                    | 0.714 [.710, .717]         | 0.881 [.878, .884]          | 0.782        | 0.793         | $R^2$  |
| <b>part <math>\eta</math></b> | <b>0.001 [−.003, .003]</b> | <b>−0.000 [−.007, .005]</b> | <b>0.220</b> | <b>0.100</b>  | $R^2$  |
| part $\phi$                   | −0.002                     | −0.005                      | −0.004       | −0.004        | $R^2$  |
| part $E$                      | 0.637 [.629, .643]         | 0.749 [.740, .757]          | 0.692        | 0.674         | $R^2$  |
| part quartile                 | 0.779 [.777, .781]         | 0.708 [.692, .728]          | 0.832        | 0.765         | Acc.   |
| part $\Delta R$               | 0.867 [.865, .869]         | 0.944 [.942, .947]          | 0.965        | 0.983         | $R^2$  |

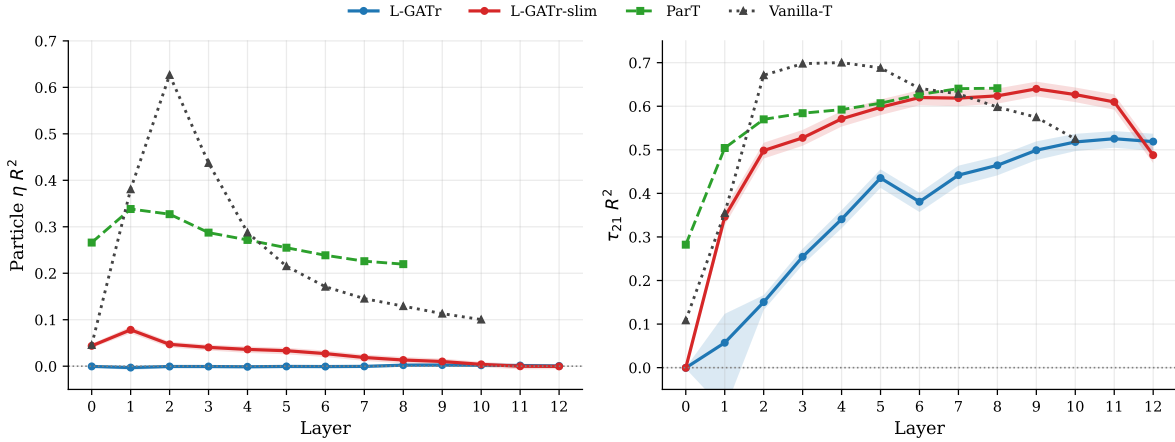


Figure 2. Two key scalar-channel probe trajectories across all four TopTagging models. Shaded bands: 95% bootstrap CIs ( $B = 1000$ ) for L-GATr and L-GATr-slim. **Left**: particle  $\eta$  is the clearest invariance diagnostic—equivariant models remain near zero throughout depth while ParT and Vanilla-T retain substantial non-zero signal. **Right**:  $\tau_{21}$  is consistently harder to decode than individual  $\tau_N$  values, with L-GATr notably weaker than ParT and Vanilla-T. Remaining probes are in Appendix C.

## 5. Structured Grade Usage in Geometric Algebra

### 5.1. Grade Decomposition: Grade-2 Is the Most Sensitive TopTagging Ablation

Table 3 reports grade decomposition results for L-GATr on TopTagging. Under this intervention, grade-2 bivectors are the most sensitive channel: zeroing all grade-2 components at every layer reduces AUC from 0.986 to 0.623 ( $\Delta\text{AUC} = 0.363$ , 95% CI [0.352, 0.373]), while zeroing any other grade degrades performance by at most 0.087. Statistical significance is confirmed at  $p < 0.001$  under  $B = 1000$  bootstrap resampling for all five grades.

The keep-only results present a complementary picture: while grade-3 trivectors alone give the highest single-grade

AUC (0.892), this does not contradict grade-2 having the largest sensitivity under the zero-grade intervention. When the full network is present, grade-3 information may already be available through grade-2 processing and the geometric product; the keep-only result instead shows grade-3 carries substantial standalone capacity when isolated. Grade-2 alone gives only 0.414, which may indicate that grade-2 encodes information that requires interaction with other grades to be fully exploited. We emphasize that both zero-grade and keep-only analyses are strong interventions on a trained network, so they should be read as sensitivity measurements rather than direct measurements of the model’s native computation.

**Physical interpretation (tentative).** Grade-2 bivectors span the Lorentz algebra  $\mathfrak{so}(1, 3)$ , comprising three boost planes

Table 3. Grade decomposition ablations for L-GATr on TopTagging (baseline AUC = 0.986,  $B = 1000$  bootstrap, all p-values < 0.001). **Zero-grade**: one grade zeroed at all layers simultaneously. **Keep-only**: all grades except one zeroed.

| GRADE              | ZERO-GRADE         |                     | KEEP-ONLY |
|--------------------|--------------------|---------------------|-----------|
|                    | $\Delta\text{AUC}$ | 95% CI              | AUC       |
| G0 SCALAR          | +0.005             | [.004, .006]        | 0.308     |
| G1 VECTOR          | +0.087             | [.081, .093]        | 0.722     |
| <b>G2 BIVECTOR</b> | <b>+0.363</b>      | <b>[.352, .373]</b> | 0.414     |
| G3 TRIVECTOR       | +0.079             | [.073, .084]        | 0.892     |
| G4 PSEUDOSCALAR    | +0.002             | [.002, .003]        | 0.691     |

$e_{0i}$  and three rotation planes  $e_{ij}$ . This could be related to the three-prong substructure of hadronic top decays ( $t \rightarrow Wb \rightarrow qq\bar{b}$ ): each pair of hard subjets defines a distinct boost/rotation plane, and grade-2 may provide a natural algebraic basis for encoding such geometric relationships. We note that this is a *post-hoc* interpretation of the empirical finding, and further study is needed to confirm it.

**L-GATr-slim contrast.** L-GATr-slim uses a different internal channel structure, and its ablation behaviour differs substantially: zeroing the scalar channels  $h_s$  reduces AUC to 0.500 (random,  $\Delta\text{AUC} = 0.486$ ), while zeroing all vector channels gives  $\Delta\text{AUC} = 0.226$ . This suggests that L-GATr-slim relies primarily on direct scalar accumulation, whereas L-GATr shows greater sensitivity to grade-2 bivector channels. Both architectures achieve the same task AUC, suggesting these may represent two distinct representational strategies for the same problem—a contrast that grade-aware analysis makes visible.

**JetClass exploratory extension.** Table 4 reports zero-grade ablations for L-GATr on JetClass (OvO AUC = 0.9884). The grade structure is strikingly different from TopTagging: grade-2 is nearly irrelevant ( $\Delta\text{AUC} = 0.023$ ), while grade-3 trivectors become dominant ( $\Delta\text{AUC} = 0.135$ ) and grade-4 pseudoscalars rise substantially ( $\Delta\text{AUC} = 0.075$  vs. 0.002 on TopTagging). We note that bootstrap confidence intervals for these ablations are not yet available and the results should be interpreted accordingly. Keep-only ablations provide a complementary view of standalone grade capacity: G0-only gives OvO AUC = 0.518, G1-only = 0.574, G2-only = 0.448, G3-only = 0.590, and G4-only = 0.546.

We treat JetClass as exploratory. Relative to TopTagging, its ablation pattern shifts toward grade-3 and grade-4 sensitivity, which may reflect the broader mix of jet topologies in the 10-class task. Cached per-class one-vs-rest AUCs sharpen this picture: zeroing grade-3 most strongly affects  $Z \rightarrow qq$  ( $\Delta\text{AUC} = 0.300$ ),  $H \rightarrow cc$  (0.242), and  $H \rightarrow bb$  (0.237), while zeroing grade-4 is most damaging for leptonic top jets ( $t \rightarrow W_{\text{lep}}b$ ,  $\Delta\text{AUC} = 0.199$ ). By contrast, zeroing grade-2

Table 4. Zero-grade ablations for L-GATr on JetClass (baseline OvO AUC = 0.9884). Keep-only runs: G0 = 0.518, G1 = 0.574, G2 = 0.448, G3 = 0.590, G4 = 0.546. Note: bootstrap CIs not yet available for JetClass;  $\Delta\text{AUC}$  values are computed from the unrounded baseline and ablated AUCs.

| GRADE               | AUC (ZEROED) | $\Delta\text{AUC}$ |
|---------------------|--------------|--------------------|
| G0 SCALAR           | 0.951        | +0.038             |
| G1 VECTOR           | 0.884        | +0.105             |
| G2 BIVECTOR         | 0.965        | +0.023             |
| <b>G3 TRIVECTOR</b> | <b>0.853</b> | <b>+0.135</b>      |
| G4 PSEUDOSCALAR     | 0.914        | +0.075             |

changes every class by at most 0.048, consistent with the small macro-level JetClass G2 effect in Table 4. These class-resolved patterns are suggestive rather than definitive because bootstrap confidence intervals are not yet available.

## 5.2. Layer-Resolved Ablation: Grade-2 Critical in Early Layers

Figure 3 shows the layer-resolved grade ablation for L-GATr on TopTagging. Grade-2 ablation causes  $\Delta\text{AUC} \approx 0.35$ –0.55 at layers 0–4, then drops sharply to approximately 0.03 at layers 5 and beyond. Grade-1 ablation decreases monotonically from layer 0 onward. Crucially, *all* grades produce  $\Delta\text{AUC} \approx 0$  at layer 12. As noted in Section 2, this is an architectural consequence of the grade-preserving `linear_out`: the output logit can only receive grade-0 signal from the final layer, regardless of what other grades encode. This is not a sign that late-layer non-G0 grades are uninformative globally—it is a direct constraint from the architecture.

The sharp drop in grade-2’s ablation impact after layer 4—combined with the monotone decrease in grade-1 ablation impact—suggests that early grade sensitivity changes substantially across depth. We explore whether this is consistent with information becoming more linearly accessible through other invariant grade combinations using multivector probes below.

## 5.3. Geometric-Algebra-Invariant Probes Trace Grade-Resolved Signals

To study how linearly decodable information is distributed across grades with depth, we fit linear probes on the 832 geometric-algebra-invariant features computed from  $h_{\text{mv}}$  at each layer (Section 3).

**Grade-2 leads at early layers.** At layer 1, the grade-2 scalar invariants ( $\langle B_i B_j \rangle_0$ ) already achieve  $R^2 = 0.492$  for jet mass, 0.551 for  $\tau_3$ , and 0.437 for  $\tau_2$ , clearly outperforming grade-1 inner products at the same layer. This suggests that grade-2 bivector channels become informative very early in the network, consistent with the grade-ablation results

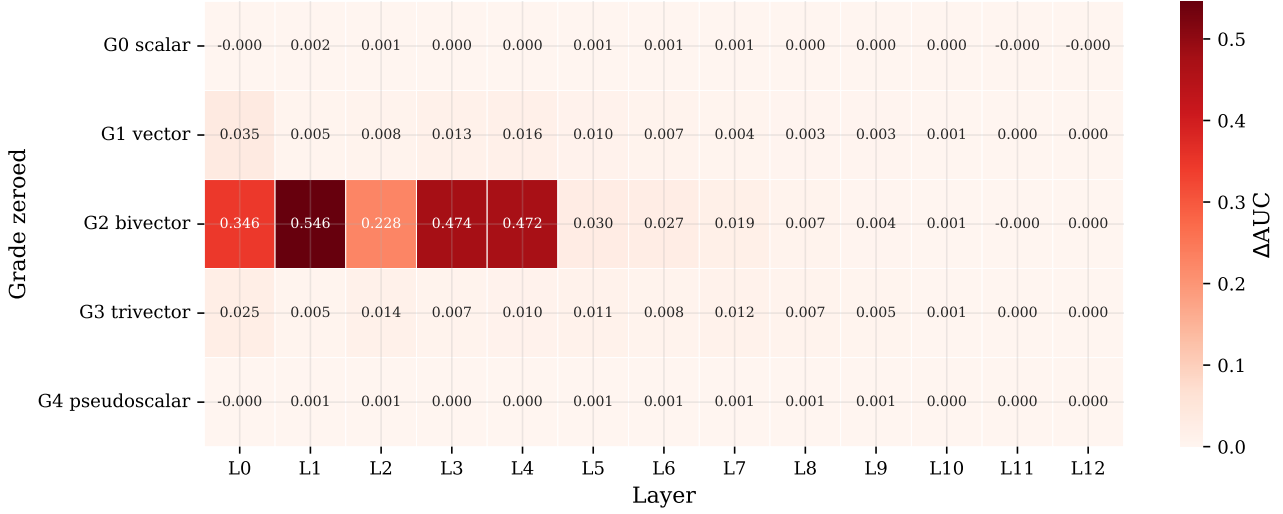


Figure 3. Layer-resolved grade decomposition for L-GATr on TopTagging. Grade-2 ablation is most impactful in early layers (L0–L4) and drops at deeper layers, consistent with a staged representational pattern in which grade-2 information becomes less directly necessary as scalar readout information accumulates. All grades produce  $\Delta\text{AUC} \approx 0$  at L12 by architectural necessity (see text).

above.

**Grade-1 and grade-3 rise in mid-layers.** By the early-to-mid network, grade-1 and grade-3 probe scores overtake grade-2 for most targets, though the exact crossover layer depends on the observable. However, the grade decomposition ablations (Section 5) show that grade-1 ablation impact also *decreases monotonically* with depth, meaning grade-1 channels at later layers are less directly important under this intervention for the final output. One possible explanation is that grade-1 and grade-3 features at deeper layers contain information correlated with earlier grade-2 processing via geometric products (e.g.  $G2 \times G1 \rightarrow G1 + G3$ ). This remains a hypothesis: the probes show linear decodability, not a complete causal pathway.

**Grade-0 saturates at the final layer.** The grade-0 direct probe achieves  $\text{AUC} = 0.986$  for top/QCD classification at layer 12, approximately matching the full model AUC. Combined with the scalar channel probe score of 0.986 at the same layer, this is consistent with the final readout concentrating discriminative information into grade-0 components of  $h_{\text{mv}}$  and  $h_{\text{s}}$ .

**Near-Hodge duality.** Grade-1 and grade-3 probe curves track each other closely for many targets, especially at later layers. In the geometric algebra of Minkowski spacetime, grade-3 trivectors are related to grade-1 vectors via the pseudoscalar  $I$  (Hodge duality). Our results may suggest that L-GATr has learned to use these two grades in a correlated fashion, though they are not identical: the cross-grade features perform slightly better than either grade alone, indicating some non-trivial inter-grade structure.

**$\tau_{21}$  and multivector structure.** While grade-2 alone encodes  $\tau_{21}$  only weakly at layer 12 ( $R^2 = 0.165$ ), grade-1, grade-3, and especially cross-grade features are stronger, and the full 832-feature probe reaches  $R^2 = 0.638$ . In the same fixed-split pipeline, the corresponding  $h_{\text{s}}$  probe reaches 0.509, so  $\tau_{21}$  appears more linearly accessible from inter-grade structure in  $h_{\text{mv}}$  than from  $h_{\text{s}}$  alone. We do not compare these fixed-split values directly to the bootstrap/OOB numbers in Table 2.

**Particle  $\eta$  in multivector probes.** Even for the 832 invariant features, the particle  $\eta$  probe peaks mid-network before declining to  $R^2 = 0.100$  at layer 12. Since these features are Lorentz-invariant by construction, this non-zero value may reflect a lab-frame correlation via the beam spurion: the beam reference token influences channel values in a way that correlates with rapidity even after taking invariant inner products. The particle  $\phi$  probe is near zero for all layers ( $R^2 \approx 0$ ), consistent with the approximate azimuthal symmetry of the dataset.

#### 5.4. Synthesis: A Staged Representational Pattern

Combining the evidence from the preceding sections, we summarise the TopTagging results as a three-stage pattern. We emphasize that this is a *descriptive* account of the observed trends rather than a direct mechanistic observation. In early layers (L0–L4), grade-2 is most important under ablation and already leads the invariant probes for mass,  $\tau_2$ , and  $\tau_3$ . In mid-layers (L5–L8), grade-2 sensitivity drops while grade-1 and grade-3 probe accessibility rises. In late layers (L9–L12), discriminative information is concentrated in grade 0, consistent with the architectural scalar readout.

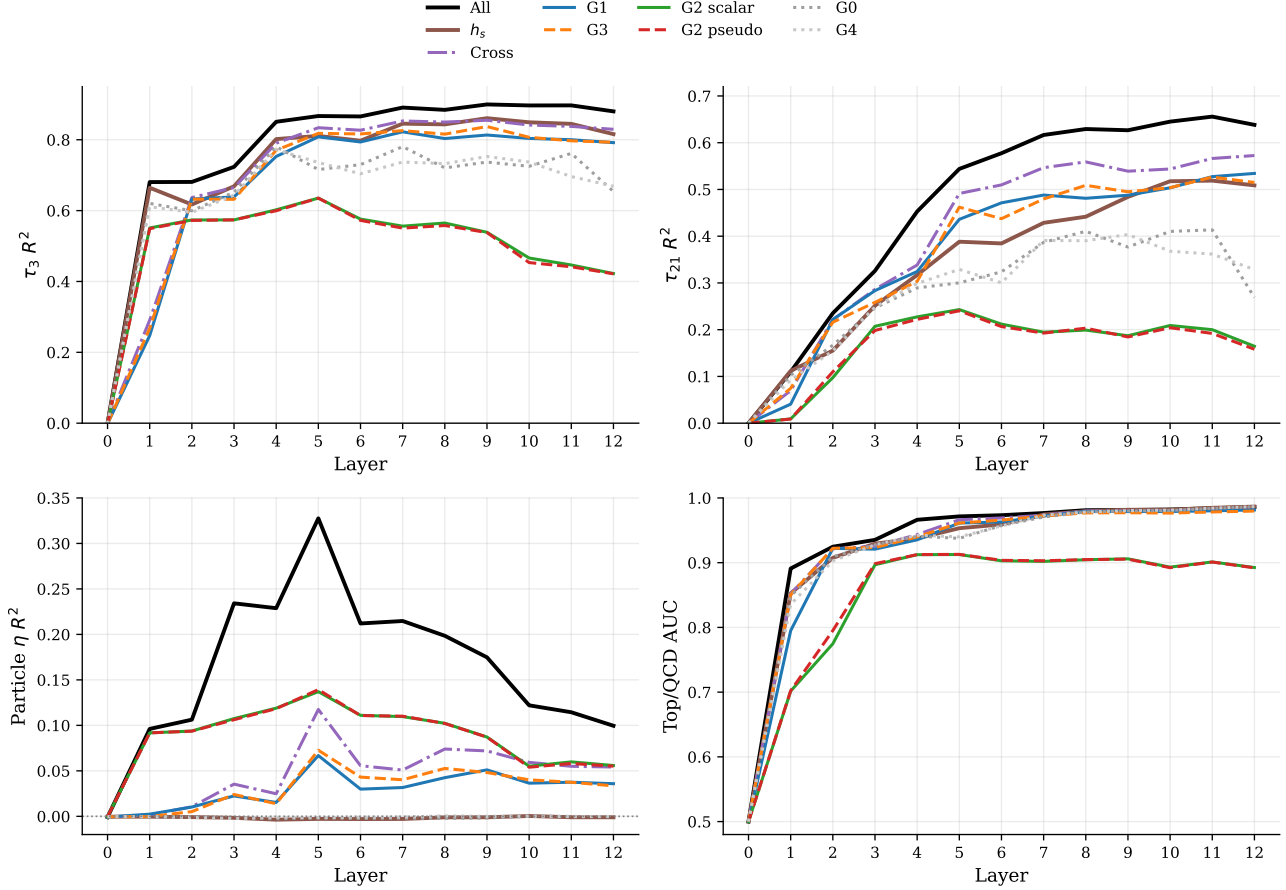


Figure 4. Geometric-algebra-invariant probe summary across four representative targets. **Top left:** for  $\tau_3$ , grade-2 leads early while grade-1, grade-3, and cross-grade features overtake it later. **Top right:** for  $\tau_{21}$ , cross-grade structure is more informative than grade-2 alone, and the full 832-feature probe is strongest within the fixed-split multivector-probe pipeline. **Bottom left:** particle  $\eta$  peaks mid-network and then declines, consistent with a residual lab-frame correlation rather than direct frame dependence in  $h_s$ . **Bottom right:** classification becomes highly linearly accessible across several invariant feature groups by later layers.

## 6. Discussion and Conclusion

We have shown that Lorentz equivariance tests, scalar-channel probes, grade decomposition, and geometric-algebra-invariant probes expose physically structured internal representations in L-GATr. The central lesson is that the model’s symmetry structure is visible inside its representations. Lorentz equivariance is not only an architectural promise: it appears in output stability under boosts and in scalar representations that suppress frame-dependent  $\eta$  information.

The grade-level analyses provide a more detailed picture. On TopTagging, grade-2 bivectors are the most sensitive ablation channel, and invariant probes show strong early grade-2 signals for jet mass and N-subjettiness. Later layers show increasing scalar accessibility, consistent with a staged representational pattern. We deliberately treat this as evidence for a hypothesis rather than a complete mecha-

nistic proof: the experiments identify where information is accessible and which interventions matter, but they do not by themselves establish every intermediate computation.

JetClass suggests that grade usage is task-dependent: the same architecture does not rely on the same grades in the same way across benchmarks. Even in this exploratory setting, that contrast strengthens the case that geometric-algebra models learn which geometric degrees of freedom matter for a given problem.

These results also suggest a practical use for symmetry-aware interpretability: auditing whether an equivariant model’s internal representations align with the symmetries and observables that matter in the domain. Grade ablations and invariant probes offer model-specific tools for asking whether an architecture has learned physically meaningful structure, rather than only improving benchmark scores.

**Limitations.** Several aspects of this study should be viewed



as early evidence rather than final mechanistic explanation. First, linear probes measure accessibility of information, not necessarily how the model uses that information internally. Second, grade ablations are interventions on trained networks and may introduce distribution shift in hidden states. Third, the physical interpretations of grade-2, grade-3, and grade-4 usage are post-hoc hypotheses that require further validation, especially through per-class JetClass studies and activation-level causal tests. Finally, our analysis focuses on existing trained checkpoints and does not yet test how these representational patterns change across seeds, model scales, or training objectives. The main TopTagging conclusions should therefore be read as findings from the analysed checkpoints rather than as seed-robust statements.

**Future directions.** First, the 832 geometric-algebra-invariant features from  $h_{mv}$  at layer 8 already achieve  $AUC = 0.981$  versus the full model’s 0.986, suggesting that early-layer invariant contractions paired with a linear head could be explored as lightweight equivariant surrogates. Second, a fuller JetClass analysis—including per-class grade structure,  $h_s$  probes across all 10 classes, and per-class attention maps—would test whether the task-dependent grade patterns persist at finer resolution. Third, adapting mechanistic interpretability methods such as activation patching, circuit analysis, and sparse autoencoders to geometric-algebra representations is an open opportunity for interpretable scientific ML.

## Impact Statement

This paper presents work whose goal is to advance the interpretability of symmetry-constrained neural networks for particle physics applications. We do not anticipate specific negative societal consequences from this work.

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## Appendix

### A. Attention Maps

Figure 5 compares mean final-layer attention maps on TopTagging for L-GATr and L-GATr-slim, averaged over approximately  $10^4$  jets and truncated to the top-50 constituents by  $p_T$ . Both models show a bright left column at position  $j = 0$ , indicating strong attention to the highest- $p_T$  constituent across many heads. L-GATr shows a sharper particle-0 focus in several heads, while L-GATr-slim displays a somewhat broader spread in some heads. The shared qualitative pattern is consistent with prior attention-map analyses of jet transformers (Wang et al., 2024; Esmail et al., 2026).

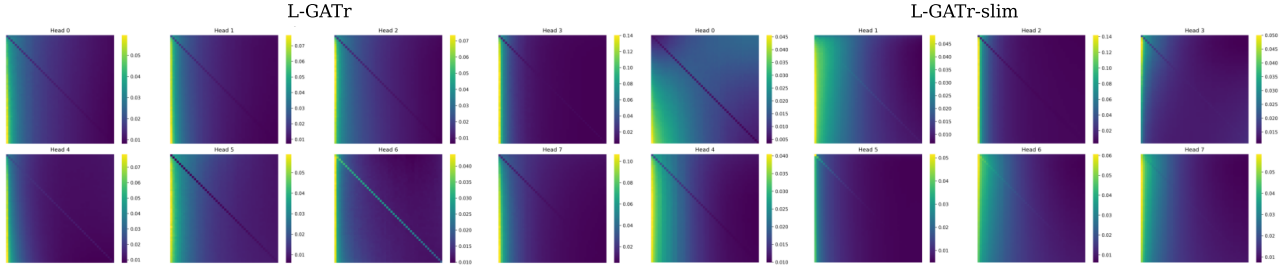


Figure 5. Mean final-layer attention maps on TopTagging for L-GATr (left, 10,013 jets) and L-GATr-slim (right, 10,003 jets), averaged over 8 heads and 50 particles. Both models strongly emphasize the leading  $p_T$  constituent, with L-GATr showing a slightly sharper particle-0 concentration in several heads.

Figure 6 shows the analogous attention maps on JetClass for both models, averaged over  $10^5$  jets. The leading- $p_T$  concentration is again visible, suggesting the pattern is not specific to the TopTagging dataset.

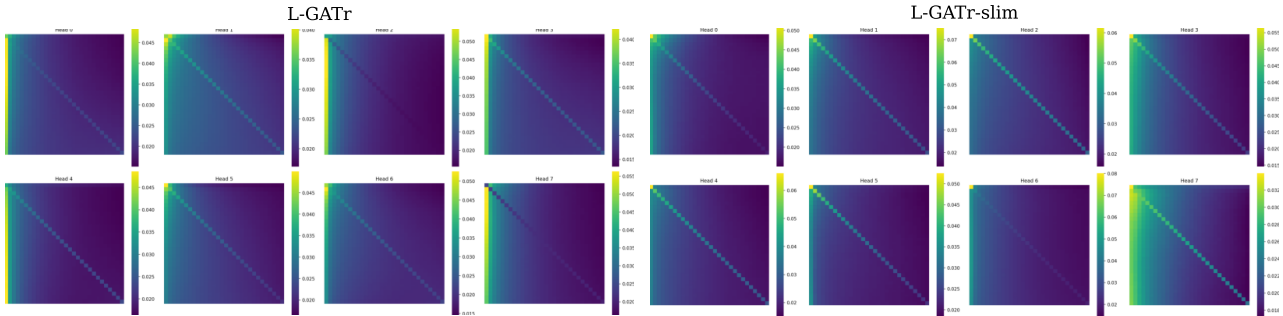


Figure 6. Mean final-layer attention maps on JetClass for L-GATr (left) and L-GATr-slim (right), averaged over  $10^5$  jets and truncated to the top-50 constituents by  $p_T$ . The qualitative pattern matches TopTagging: both models concentrate attention on the leading- $p_T$  constituent, suggesting this is a recurring feature of the learned representations.

### B. CKA Representation Similarity

We compute linear Centered Kernel Alignment (CKA (Kornblith et al., 2019)) on scalar-channel representations as a supporting diagnostic for how representations evolve with depth. Unlike the probe and ablation analyses in the main text, these plots are descriptive: they show where layers become similar or reorganise, but they do not by themselves identify which physics observables are encoded.

**TopTagging CKA.** For TopTagging, we compute linear CKA between scalar channel representations  $h_s$  at all pairs of layers, for both L-GATr and L-GATr-slim (5,000 jets, optimised Gram matrix approach). We exclude layer 0 (the output of the input linear projection) from the CKA matrices: the global token at that stage carries no jet-specific information—it is a fixed learned embedding that has not yet attended to any particles—so its representation is constant across jets and its CKA with every other layer is identically zero. The resulting  $12 \times 12$  matrices (layers L1–L12) characterise the degree to which different layers develop similar representations.

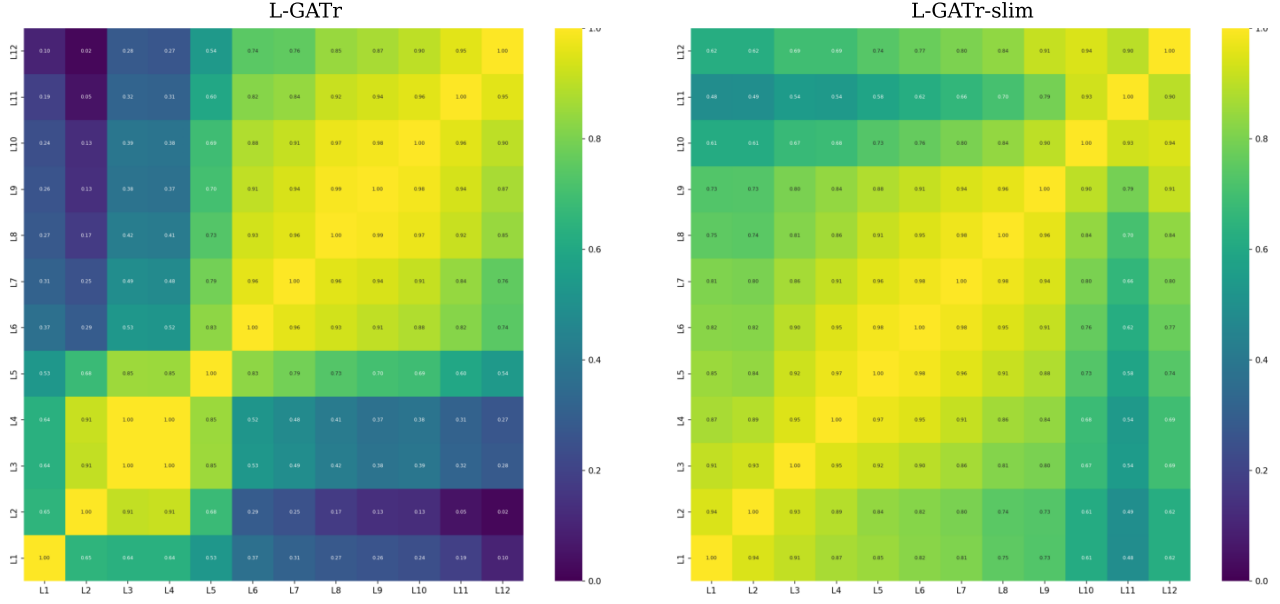


Figure 7. Linear CKA between global-token scalar representations at all layer pairs (L1-L12) for L-GATr (left) and L-GATr-slim (right) on TopTagging (5,000 jets). Layer 0 is excluded as the global token there is a constant embedding with no jet-specific information. L-GATr-slim shows a broader high-similarity band across depth, while L-GATr separates more clearly into early and late representational regimes.

**JetClass CKA.** Figure 8 shows the corresponding CKA matrices on JetClass. Unlike the TopTagging case, layer 0 is retained here because the JetClass models use mean-pooled particle representations (not a dedicated global token), so the L0 representation does vary across jets.

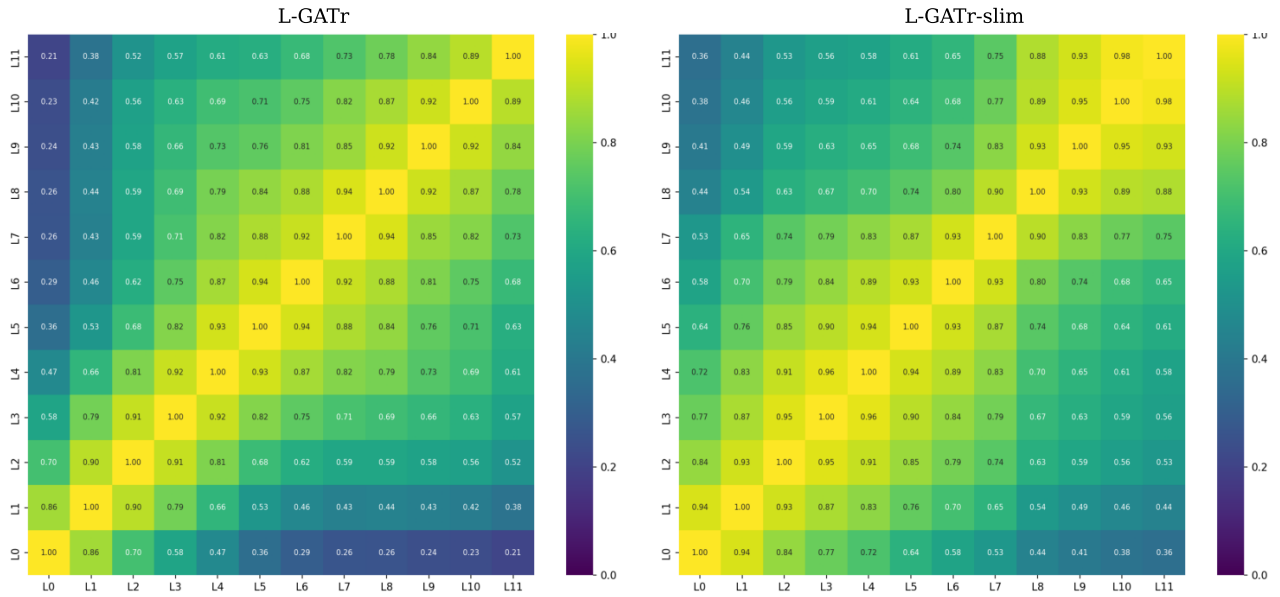


Figure 8. Linear CKA between mean-pooled scalar representations at all layer pairs (L0-L11) for L-GATr (left) and L-GATr-slim (right) on JetClass (5,000 jets). Both models show a clear block structure: early layers (L0-L4) form one high-similarity cluster and late layers (L6-L11) form another, with a representational transition around L5.

## C. Scalar Probes and Bootstrap Curves

This section shows the remaining  $h_s$  probe trajectories not included in the main-paper summary. For L-GATr and L-GATr-slim, shaded bands denote 95% bootstrap confidence intervals ( $B = 1000$ , OOB design); ParT and Vanilla-T are shown as point estimates.

**Probe training details.** For the TopTagging  $h_s$  probes, we work with 10,000 jets and fit probes at every layer. For L-GATr and L-GATr-slim, the uncertainty bands are produced by  $B = 1000$  bootstrap resamples: each resample is drawn at the jet level, the probe is fit on the in-bag jets, and evaluation is performed on the corresponding out-of-bag jets. This provides strict held-out scoring for each bootstrap replicate. Regression targets use Ridge regression with  $\alpha = 1.0$ , while binary and multi-class targets use logistic regression with L2 regularization ('lbfgs',  $C = 1.0$ ). For the fixed-split reference pipelines, features are standardized before probe fitting. For particle-level targets, the train/test split is still defined at the jet level and all particles belonging to a training jet stay on the training side, preventing particle leakage across splits; the bootstrap pipeline uses all particles associated with the selected jets. We use 10,000 TopTagging jets in these analyses as a tractable subset that still supports stable per-layer probe estimation and bootstrap uncertainty computation across many targets.

**Ablation and significance details.** For the TopTagging grade-decomposition experiments, zero-grade ablation means zeroing the selected multivector grade at the outputs of all 13 network stages (the input linear layer plus 12 transformer blocks). Keep-only ablation means zeroing all other grades at those same hidden outputs. Layer-resolved ablation zeros one grade at exactly one stage output and leaves all others untouched. These hooks operate on hidden multivectors rather than on the raw input embedding. The TopTagging confidence intervals and p-values are obtained from a paired bootstrap over jets using saved per-jet logits: each resample computes  $\Delta\text{AUC} = \text{AUC}(\text{baseline}) - \text{AUC}(\text{ablated})$ , the reported 95% interval is the 2.5th–97.5th percentile of those bootstrap  $\Delta\text{AUC}$  values, and the one-sided p-value is the fraction of resamples with  $\Delta\text{AUC} \leq 0$ .

**Reference pipelines used in the tables.** Table 2 in the main text reports the bootstrap/OOB probe scores for L-GATr and L-GATr-slim, together with fixed-split reference scores for ParT and Vanilla-T. By contrast, the  $h_s$  column in the multivector appendix tables below is included only as a same-split reference within the multivector-probe pipeline. Those appendix  $h_s$  values are fit with the fixed train/test protocol used for the multivector analysis and therefore are not expected to match the bootstrap/OOB values in Table 2 exactly.

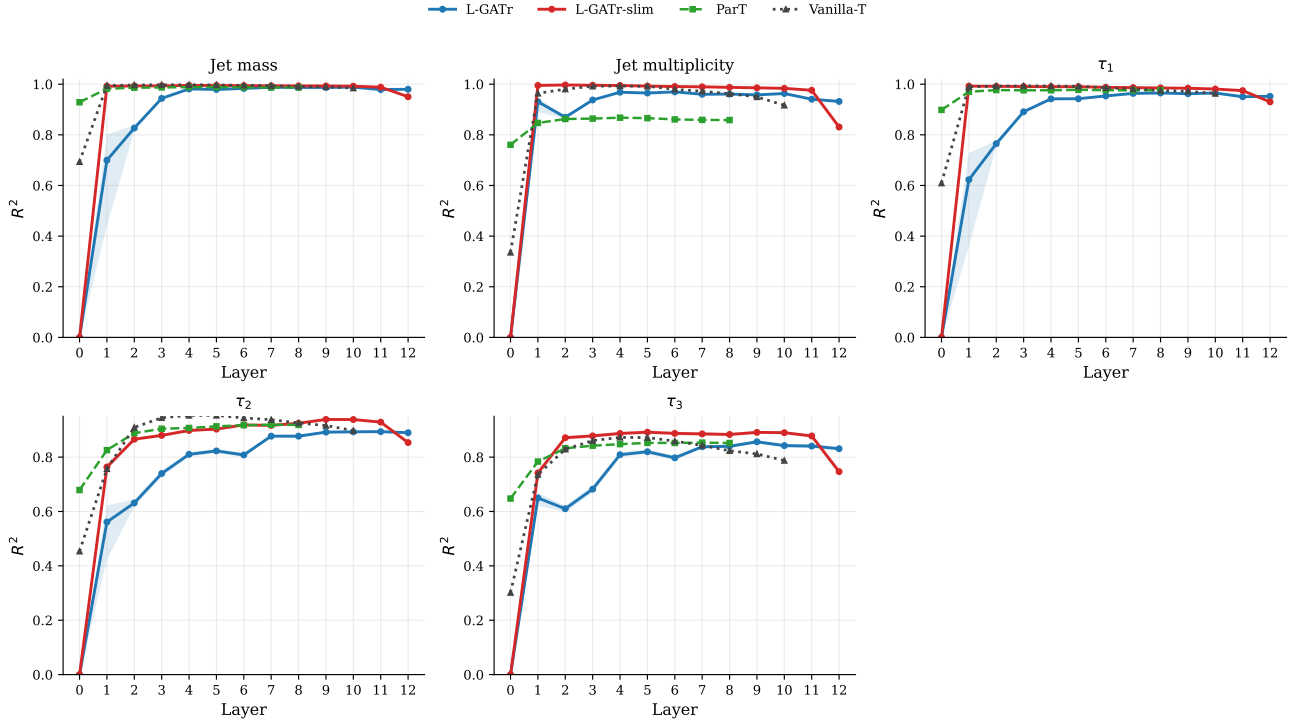


Figure 9. Remaining jet-level  $h_s$  probe curves across all four TopTagging models: jet mass, jet multiplicity,  $\tau_1$ ,  $\tau_2$ , and  $\tau_3$ .



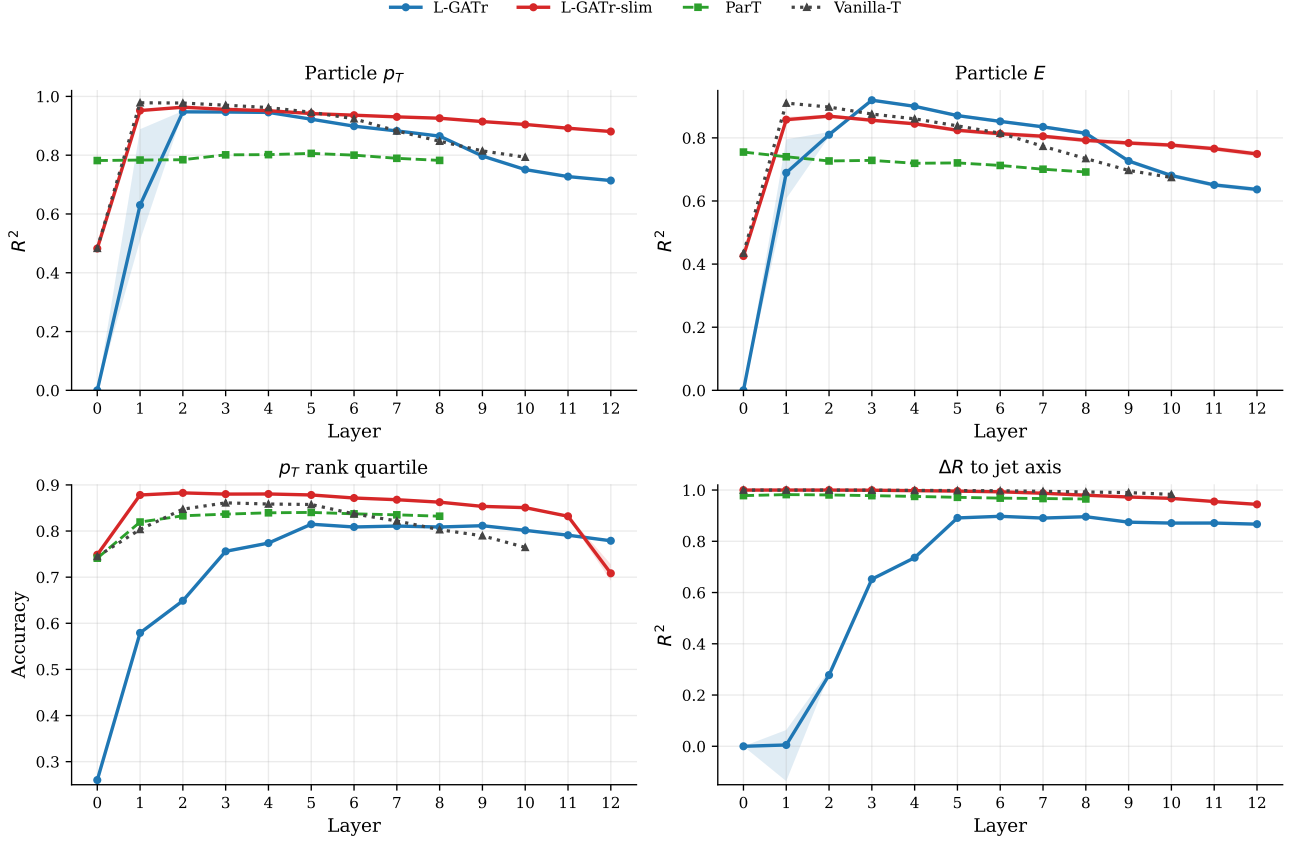


Figure 10. Remaining particle-level  $h_s$  probe curves across all four TopTagging models: particle  $p_T$ , particle  $E$ ,  $p_T$ -rank quartile, and particle  $\Delta R$ .

#### D. L-GATr-slim Grade Decomposition on TopTagging

L-GATr-slim uses a different internal channel structure from L-GATr: it retains only grade-0 scalars ( $h_s$ ) and grade-1 4-vectors ( $h_v$ ), and its readout is the scalar channels at a global token. Table 5 reports channel ablations for L-GATr-slim on TopTagging. The contrast with L-GATr (Table 3) is striking: zeroing all scalar channels reduces L-GATr-slim’s AUC to 0.500 (random,  $\Delta\text{AUC} = 0.486$ ), while the analogous zeroing of grade-2 bivectors in L-GATr gives  $\Delta\text{AUC} = 0.363$ . L-GATr-slim’s scalar channels carry the full discriminative load because its readout is *directly* the global scalar token—there is no grade-0 “distillation” step since grade-0 is the only output channel. The vector channels matter significantly less ( $\Delta\text{AUC} = 0.226$ ) but are far from negligible: they provide the geometric structure from which scalars are computed via the Minkowski inner product in each block.

The component-level ablations reveal which 4-vector components matter most: the energy component  $e_t$  is most critical ( $\Delta\text{AUC} = 0.408$ ), followed by the beam-axis component  $e_z$  ( $\Delta\text{AUC} = 0.304$ ). The transverse components  $e_x, e_y$  together give  $\Delta\text{AUC} = 0.323$ , similar to  $e_z$  alone. This hierarchy may be consistent with the physical intuition that longitudinal momentum and energy (which together determine the rapidity of a jet) are more discriminative for TopTagging than the azimuthal structure.

Table 5. Channel ablations for L-GATr-slim on TopTagging (baseline AUC = 0.986). Bootstrap CIs ( $B = 1000$ ) available for the three main ablations; all p-values < 0.001.

| CHANNEL ZEROED               | AUC          | $\Delta$ AUC  | 95% CI       |
|------------------------------|--------------|---------------|--------------|
| $h_s$ (ALL SCALARS)          | <b>0.500</b> | <b>+0.486</b> | [.485, .488] |
| $h_v$ (ALL VECTORS)          | 0.760        | +0.226        | [.224, .229] |
| $e_t$ (ENERGY COMPONENT)     | 0.579        | +0.408        | [.407, .409] |
| $e_z$ (BEAM AXIS)            | 0.683        | +0.304        | —            |
| $e_x, e_y$ (TRANSVERSE)      | 0.663        | +0.323        | —            |
| $e_x, e_y, e_z$ (3-MOMENTUM) | 0.654        | +0.333        | —            |

## E. Full Multivector Invariant Probe Results

Tables 6 and 7 report linear probe scores for all 832 Lorentz-invariant features at the final layer (L12), broken down by grade group. The features are described in Section 3; they are invariant by construction, so any predictive power they carry reflects physics encoding in the multivector channels up to any lab-frame correlations introduced through beam-spurion conditioning. The  $h_s$  column in these tables is included as a within-pipeline reference from the same fixed-split multivector analysis and should not be read as numerically identical to the bootstrap/OOB values in the main-text scalar-probe table.

**Invariant-feature details.** For the  $h_{mv}$  analysis, the 832 invariant features are grouped into direct grade-0/4 components, grade-1 pairwise inner products, grade-3 pairwise inner products, grade-1  $\times$  grade-3 cross terms, and grade-2 scalar and pseudoscalar contractions. Jet-level probes use a fixed 8k/2k train/test split over the 10,000 jets, using the jet-level multivector representation of the global jet token in the multivector stream. Particle-level probes use a random sample of 50,000 particles with an 80/20 split inside that sample. These probes use feature standardization, Ridge regression with  $\alpha = 10.0$  for regression targets, and logistic regression with ‘lbfgs’ and  $C = 0.1$  for classification targets. The tables in this section report these same-split results so that grade groups can be compared directly within one protocol. Throughout this section, dashes indicate grade/target combinations that were not run or are not meaningful for that feature family.

**Jet-level probes (Table 6).** The “All” column (all 832 features combined) matches or exceeds the scalar channel  $h_s$  probe for every jet-level target shown here, and exceeds it clearly for jet multiplicity and  $\tau_1$  as well. For  $\tau_{21}$ , the full invariant set (All = 0.638) notably exceeds  $h_s$  (0.509) by a substantial margin within this same fixed-split probe pipeline, suggesting that  $\tau_{21}$  information is encoded in the inter-channel grade structure of  $h_{mv}$  but is less linearly accessible from  $h_s$  alone under this protocol. Grade-0 (G0) and grade-4 (G4) direct components perform near each other and are surprisingly competitive with the full probe for classification (G0 = 0.986, G4 = 0.987), confirming that the model concentrates discriminative invariants directly into the grade-0/4 blade values as well as distributing them across inter-channel products. Grade-2 scalar and pseudoscalar invariants ( $\langle B_i B_j \rangle_0$  and  $\langle B_i B_j \rangle_4$ ) track each other closely—the parity-odd pseudoscalar contributes roughly equally to the parity-even scalar, consistent with QCD’s parity symmetry at the ensemble level.

Table 6. Geometric-algebra-invariant probe scores at the final layer (L12) for L-GATr on TopTagging, broken down by grade group and target. Metric:  $R^2$  for regression, AUC for classification (label row). The  $h_s$  column is the same-split scalar reference from the multivector probe pipeline, not the bootstrap/OOB estimate from Table 2. Dashes: probes not run for those grade/target combinations.

| Target          | $h_s$ | G0    | G4    | G1    | G3    | G1 $\times$ G3 | G2 scalar | G2 pseudo | All   |
|-----------------|-------|-------|-------|-------|-------|----------------|-----------|-----------|-------|
| top/QCD         | 0.987 | 0.986 | 0.987 | 0.981 | 0.980 | 0.982          | 0.893     | 0.892     | 0.986 |
| jet mass        | 0.976 | 0.916 | 0.941 | 0.958 | 0.956 | 0.973          | 0.767     | 0.762     | 0.988 |
| jet mult        | 0.902 | —     | —     | 0.806 | 0.862 | —              | 0.421     | 0.429     | 0.962 |
| jet $\tau_1$    | 0.950 | —     | —     | 0.948 | 0.944 | —              | 0.719     | 0.713     | 0.972 |
| jet $\tau_2$    | 0.881 | 0.782 | 0.793 | 0.872 | 0.866 | 0.887          | 0.546     | 0.543     | 0.917 |
| jet $\tau_3$    | 0.816 | 0.654 | 0.669 | 0.792 | 0.793 | 0.830          | 0.422     | 0.421     | 0.880 |
| jet $\tau_{21}$ | 0.509 | 0.269 | 0.329 | 0.534 | 0.515 | 0.573          | 0.165     | 0.158     | 0.638 |

**Particle-level probes (Table 7).** The  $\eta$  probe for the invariant features (All = 0.100) is non-zero despite these features being Lorentz-invariant by construction. As discussed in Section 5.3, this residual correlation is a lab-frame effect mediated by the beam spurion rather than a failure of invariance. Grade-2 invariants (G2 scalar = 0.056) are the most predictive of  $\eta$  among individual grade groups—consistent with bivector boost planes ( $e_{0i}$ ) encoding rapidity-like information via spurion-influenced channels. The  $\phi$  probe is near zero for all groups ( $\lesssim 0.006$ ), confirming the azimuthal sanity check. For

$p_T$  and  $E$ , the “All” features substantially exceed any single grade group, indicating that momentum encoding is distributed across the full grade structure of  $h_{mv}$ .

Table 7. Particle-level geometric-algebra-invariant probe scores at the final layer (L12) for L-GATr on TopTagging. Metric:  $R^2$  for regression, accuracy for part quartile. The  $\eta$  row is non-zero due to lab-frame beam spurion correlations, not a violation of Lorentz invariance (see text). The  $h_s$  column is the same-split scalar reference from the multivector probe pipeline, not the bootstrap/OOB estimate from Table 2.

| Target                        | $h_s$         | G1           | G2 scalar    | G2 pseudo    | G3           | All          |
|-------------------------------|---------------|--------------|--------------|--------------|--------------|--------------|
| part $p_T$                    | 0.943         | 0.908        | 0.488        | 0.474        | 0.906        | 0.970        |
| <b>part <math>\eta</math></b> | <b>-0.001</b> | <b>0.036</b> | <b>0.056</b> | <b>0.055</b> | <b>0.034</b> | <b>0.100</b> |
| part $\phi$                   | -0.000        | -0.002       | -0.001       | -0.001       | -0.002       | -0.006       |
| part $E$                      | 0.928         | 0.868        | 0.461        | 0.449        | 0.866        | 0.964        |
| part quartile                 | 0.802         | 0.801        | 0.355        | 0.347        | 0.791        | 0.870        |
| part $\Delta R$               | 0.867         | 0.837        | 0.563        | 0.540        | 0.884        | 0.948        |

Figure 11 shows the remaining per-layer multivector invariant probe trajectories not included in the main-paper summary.

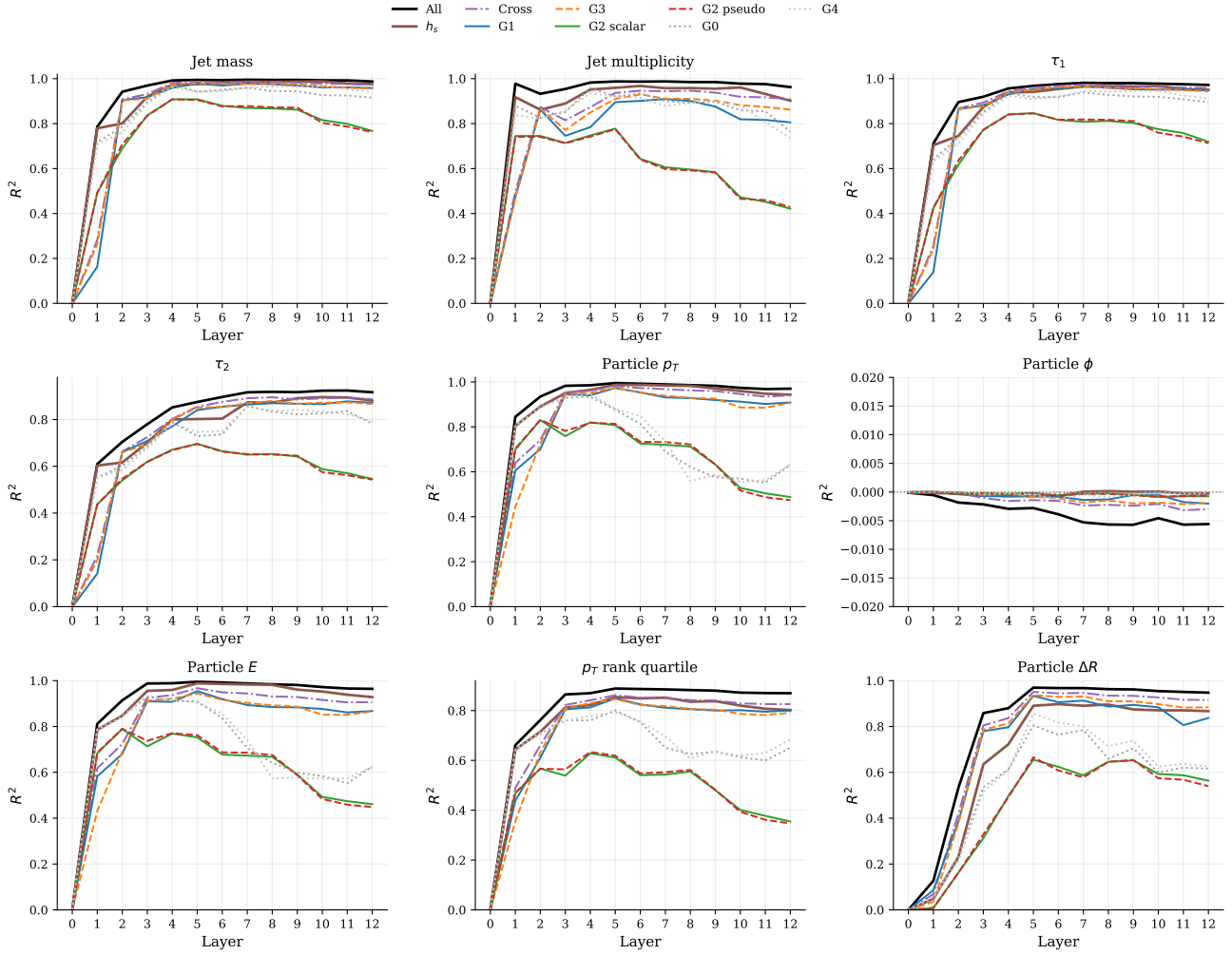


Figure 11. Remaining per-layer geometric-algebra-invariant probe curves, broken down by grade group: jet mass, jet multiplicity,  $\tau_1$ ,  $\tau_2$ , particle  $p_T$ , particle  $\phi$ , particle  $E$ ,  $p_T$ -rank quartile, and particle  $\Delta R$ .

## F. Per-Layer Equivariance Details

Table 8 reports the per-layer equivariance errors for L-GATr on TopTagging (200 jets, 5 random Lorentz transforms each). We measure two quantities: the relative change in  $h_s$  (which should be invariant, i.e. error = 0) and the relative change in  $h_{mv}$  (which should transform equivariantly, i.e. error  $\neq 0$  but substantially below the naïve no-change baseline of  $\sim 1.3$ ).

**Equivariance metric details.** For the hidden-state checks, scalar-channel invariance and multivector/vector equivariance are measured with relative Frobenius errors of the form  $\|a - b\|_F / (\|b\|_F + \varepsilon)$  with  $\varepsilon = 10^{-8}$ . For the output-level summary in the main text, the reported logit change is  $|\ell(\Lambda x) - \ell(x)| / (|\ell(x)| + \varepsilon)$ . For full L-GATr, the expected transformed multivectors use the Pin/Spin representation  $\rho(\Lambda)$ ; for L-GATr-slim, the expected transformed hidden vectors use the corresponding  $4 \times 4$  Lorentz matrix. In the implemented tests, the Lorentz transformation is applied to the full embedded input state, including the particle 4-momenta and the spurion/reference tokens.

At layer 0 (`linear_in`),  $h_s$  invariance is exact to machine precision (0), and  $h_{mv}$  equivariance error is  $5 \times 10^{-8}$  (float32 machine epsilon)—`linear_in` is exactly equivariant. Across layers 1–12, errors accumulate to  $10^{-4}$ – $5 \times 10^{-3}$  for  $h_s$  and  $7 \times 10^{-3}$ – $10^{-2}$  for  $h_{mv}$ , consistent with floating-point drift across 12 GATrBlock operations. The naïve baseline (treating a boosted input as unchanged) gives errors of  $\sim 1.3$  throughout, confirming that the equivariant architecture reduces the effective error by approximately  $100\times$  at all layers.

Table 8. Per-layer equivariance errors for L-GATr on TopTagging (200 jets  $\times$  5 random Lorentz transforms).  $h_s$  error: relative change in scalar channels (should be 0).  $h_{mv}$  equivariance: relative change after applying the correct group action. Naïve: error if no transformation applied (upper bound baseline).

| Layer                         | $h_s$ error                    | $h_{mv}$ equiv.                | Naïve      |
|-------------------------------|--------------------------------|--------------------------------|------------|
| L0 ( <code>linear_in</code> ) | 0 (exact)                      | $5 \times 10^{-8}$             | 1.36       |
| L1–L12                        | $10^{-4}$ – $5 \times 10^{-3}$ | $7 \times 10^{-3}$ – $10^{-2}$ | $\sim 1.3$ |
| Output logit                  | $5.61 \times 10^{-3}$ (0.56%)  |                                |            |

Table 9 reports the boost sweep for L-GATr, applying pure Lorentz boosts in a random direction at increasing boost factors  $\gamma$ . Logit errors remain below 1% for  $\gamma \leq 2$  (the physically relevant regime for LHC top jet production) and grow at larger  $\gamma$  due to floating-point precision breakdown at extreme boosts, not a fundamental violation of equivariance.

Table 9. Boost sweep for L-GATr on TopTagging (pure boosts in a random direction, 50 jets).  $h_{mv}$  error: relative change in multivector hidden states. Logit error: relative change in output logit.

| $\gamma$ | $h_{mv}$ ERROR       | LOGIT ERROR          |
|----------|----------------------|----------------------|
| 1.5      | $4.5 \times 10^{-3}$ | $2.3 \times 10^{-3}$ |
| 2.0      | $1.0 \times 10^{-2}$ | $7.7 \times 10^{-3}$ |
| 5.0      | $4.0 \times 10^{-2}$ | $4.6 \times 10^{-2}$ |
| 10.0     | $9.5 \times 10^{-2}$ | $9.8 \times 10^{-2}$ |
| 50.0     | $4.0 \times 10^{-1}$ | 1.61                 |